

Sources Sought Notice

PIT Tag Detection Equipment

Procurement Request (PR) PR0044038137

Department of the Interior
United States Geological Survey (USGS)
Western Fisheries Research Center (WFRC)

The Department of the Interior, United States Geological Survey (USGS), Western Fisheries Research Center (WFRC) is conducting market research to identify qualified small businesses capable of supplying submersible Passive Integrated Transponder (PIT) tag detection equipment and associated components supporting fisheries monitoring operations.

This Sources Sought notice is issued for market research purposes only in accordance with Federal Acquisition Regulation (FAR) Part 10. This notice does not constitute a solicitation, Request for Quote (RFQ), or Request for Proposal (RFP), and no contract or purchase order will be awarded as a result of this announcement.

Responses are requested from small business concerns capable of providing equipment meeting the technical requirements summarized below. Responses should be concise and directly address the requested information. Marketing materials not specific to the required equipment are discouraged.

The Government will not pay for information requested nor compensate respondents for any cost incurred in developing or submitting responses. Materials submitted will not be returned.

NAICS Code

334511 — Search, Detection, Navigation, Guidance, Aeronautical, and Nautical System and Instrument Manufacturing

Size Standard: 1,350 Employees

PSC Code: 2060 – Commercial Fishing Equipment

Requirement Summary

USGS requires procurement of submersible PIT tag detection equipment to support monitoring of PIT-tagged fish during seasonal migration studies. Equipment must operate autonomously in aquatic environments and integrate with existing PIT detection monitoring infrastructure.

Equipment to be supplied includes:

- Submersible rigid PIT tag antennas
- Receiver nodes and associated detection electronics
- Submersible battery systems and charging equipment
- Associated components required for deployment and operation

Equipment must be capable of long-duration deployment in aquatic environments and support reliable PIT tag detection and data storage during fisheries monitoring operations. REFERENCE INCLUDED TECHNICAL SPECIFICATIONS.

Requested Vendor Information

(Capability Statement – Maximum 3 Pages)

Interested small businesses are requested to submit capability statements addressing the following:

1. Experience supplying PIT tag detection or comparable fisheries monitoring equipment.

2. Ability to supply submersible PIT detection antennas, receivers, and battery systems compatible with existing deployed PIT detection infrastructure.
3. Description of equipment meeting operational requirements for submerged deployment environments.
4. Manufacturing and delivery lead times for equipment.
5. Available technical support, warranty support, and post-delivery services.
6. Company UEI number and confirmation of active SAM registration.
7. Business size and socioeconomic classification under NAICS 334511.
8. Identification of equipment manufacturer if respondent is a distributor or integrator.

Respondents should not reply if unable to supply equipment compatible with deployed PIT detection infrastructure.

Submission Instructions

Responses must be submitted electronically via email to:

Joel Berberena
Contracting Officer
Email: jberberena@usgs.gov

Subject Line:

Sources Sought Response – PIT Tag Detection Equipment

Notice

This notice is issued solely for market research purposes and does not constitute a solicitation or commitment by the Government. Information received will be used to determine acquisition strategy, including potential small business set-aside determinations.

1. Acoustic Transmitters (Tags)

Transmitters shall meet the following requirements:

Weight in air between 3.5 and 3.7 grams.

Binary Phase Shift Keyed (BPSK) coded pulse signals enabling unique identification.

Available in size configurations allowing maximum battery life selection while maintaining a tag-to-body weight ratio of 2 percent or less.

Suitable for fish sizes of 350 grams or greater, depending on transmitter selected.

Operating temperature range of 0–40 °C.

Operating frequency of 69 kHz $\pm$ 0.5%.

Acoustic output power of at least 146 dB.

Optional integrated temperature sensor operating from 0–40 °C with $\pm$ 0.5 °C accuracy.

2. Acoustic Receivers

Receivers shall meet the following requirements:

Detect, decode, and store tag detections from 69 kHz BPSK acoustic transmitters.

Internal memory capable of storing millions of tag detection records.

Autonomous operation powered by internal battery.

Fully waterproof to depths of at least 700 meters.

Bluetooth connectivity enabling receiver ID configuration, clock synchronization, configuration access, and viewing of real-time data via vendor-provided graphical software.

Record system diagnostics including noise, receiver tilt, temperature, and signal strength.

Include programmable built-in synchronization transmitter.

Operating temperature range of 0–50 °C.

Minimum one-year parts and labor warranty.

3. Acoustic Receiver Activation Probe

Activation probes shall:

Enable activation of receiver Bluetooth interface.

Allow configuration access including receiver ID setup, clock synchronization, and real-time data viewing through vendor software.

Operate within a temperature range of 0–50 °C.

Include minimum one-year parts and labor warranty.

4. PIT Tag Receiver System

Receivers shall meet the following requirements:

Antenna Requirements

5-foot diameter rigid submersible PIT-tag antenna.

Tag Compatibility

Detection of ISO-compliant FDX-B and HDX PIT tags.

Operating frequency of 134.2 kHz.

U.S. Department of the Interior – U.S. Geological Survey
Technical Specifications Attachment
Procurement Request (PR): PR0044038137
Acoustic & PIT Tag Detection Equipment Specifications

Receiver Performance

Autonomous, fully waterproof receiver construction.
Internal battery capacity supporting more than 30 days of continuous operation.
Internal storage capacity for at least 100,000 tag IDs.

Communications & Configuration

Bluetooth connectivity enabling receiver ID configuration, clock synchronization, configuration access, and real-time data viewing through vendor-provided software.

Environmental Requirements

Operating temperature range of –20 to 70 °C.

Warranty

Minimum one-year parts and labor warranty.

END TECHNICAL SPECIFICATIONS